

Data set -- CO2 emissions

Dataset Overview

Global Emissions Data: 198 Countries

Dimension	Details
Size	43,746 observations • 80 features
Emissions	CO ₂ , CH ₄ , N ₂ O, total GHG • Coal, oil, gas, cement
Economics	GDP, population, per-capita rates
Period	Historical through 2023

Finding #1: Country Emission Clusters

t-SNE & Clustering Methodology

Step 1 — t-SNE (Pattern Discovery)

- Input: 76 standardized emission features per country
- Output: 2D map (each point = one country)
- Countries close together = similar emission profiles

Step 2 — K-Means (Profile Groups)

- Input: 2D t-SNE coordinates (198 countries)
- Elbow method → **k** = 4 clusters
- Output: 4 distinct emission profile groups

Cluster Characteristics

Cluster	Countries	Pop (M)	GDP (\$B)	CO ₂ (MT)	CO ₂ pc	CH ₄	GHG
0	45	1.4	13.3	249.6	4.31	7.5	34.6
1	59	39.1	350.0	186.1	3.77	56.0	284.9
2	61	13.1	79.1	161.3	1.12	17.8	63.0
3	33	0.4	2.4	448.5	6.71	62.5	355.3

Cluster Profiles

Cluster	Profile	Key Stats	Examples	Policy
0	Small Nations	1.4M pop, 4.31 t/pc	Bhutan, Botswana	Green development
1	Developed	39M pop, 3.77 t/pc (efficient)	Australia, Austria	Carbon pricing
2	Developing	1.12 t/pc (lowest)	Bangladesh, Algeria	Tech transfer
3	Micro-States	6.71 t/pc (highest)	Barbados, Andorra	Sustainable tourism

Finding #2: Dangerous Future Trajectories

Forecasting Methodology

Data

- World-level annual series
- 6 variables: CO₂, total GHG, CH₄, N₂O, coal CO₂, cement CO₂
- Last 20 years shown for context

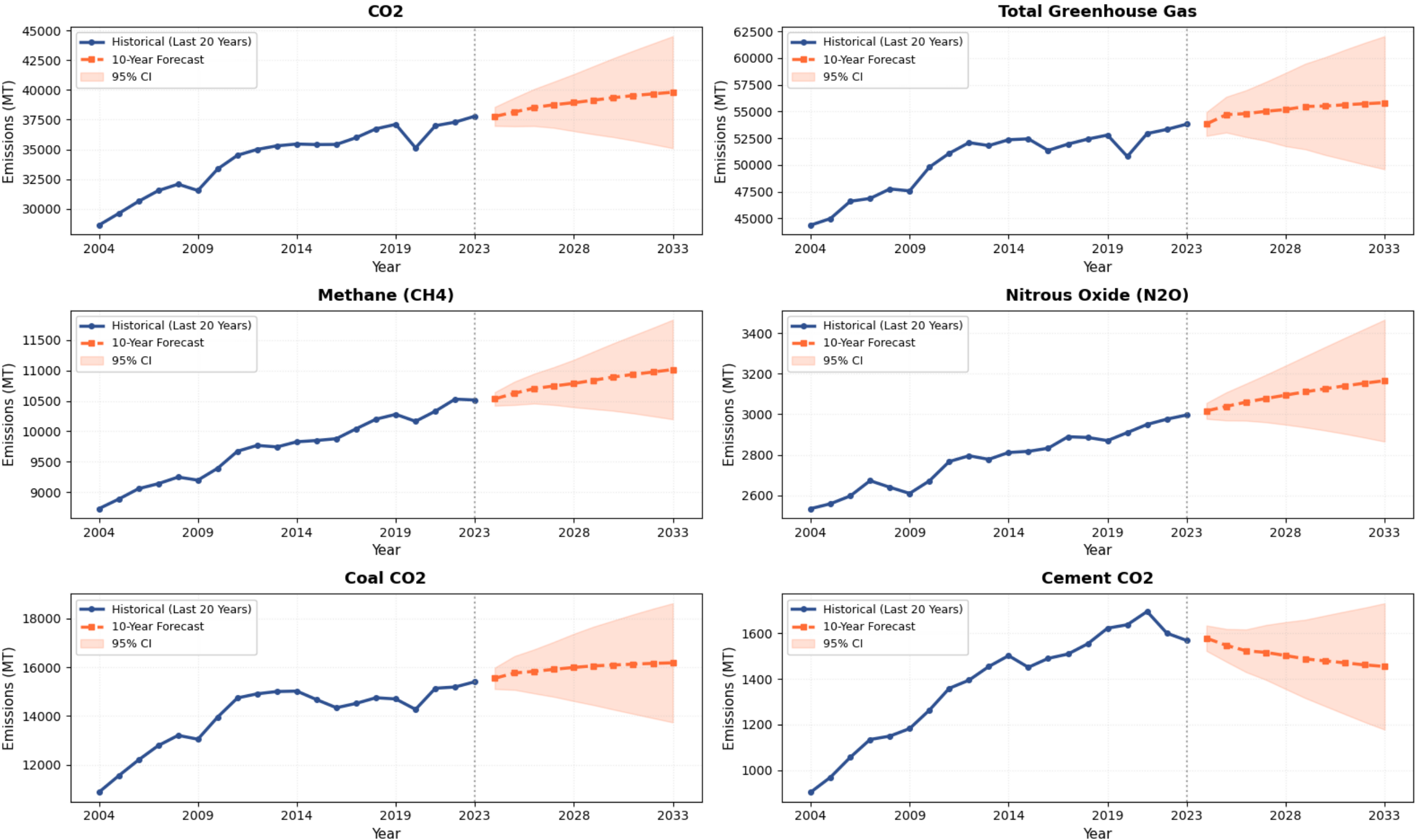
Model: ARIMA(5,1,0)

- 5 AR lags, 1 differencing, 0 MA
- 10-year forecast horizon
- 95% confidence intervals

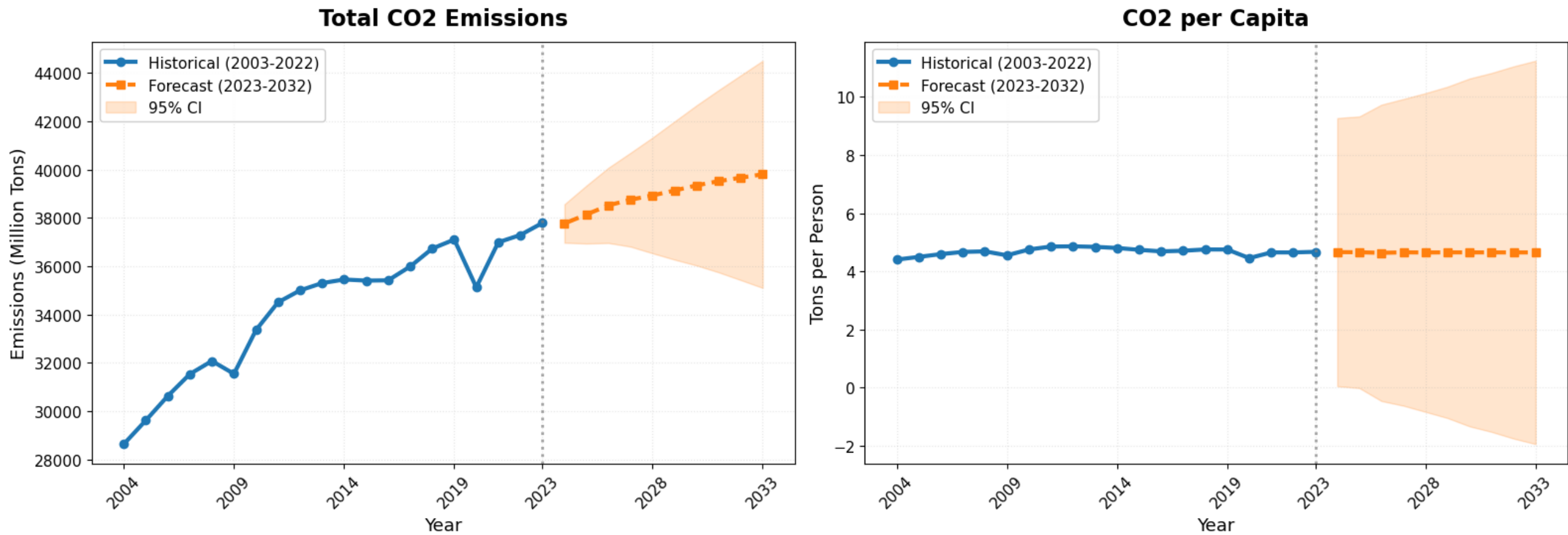
Purpose

- Compare total CO₂ vs CO₂ per capita
- Detect population vs per-capita drivers
- Early-warning tool for policy triggers

Global Emission Forecasts: 20-Year Historical Context + 10-Year Projection (ARIMA 5,1,0)



CO2 Emissions: Total vs. Per Capita - 20-Year Context + 10-Year Forecast



Finding #3: Economic- Environmental Coupling

Correlation Methodology

Analysis

- Pearson correlation matrix
- Identify strongest relationships

Purpose

- Quantify GDP-emission linkage
- Evidence of structural coupling

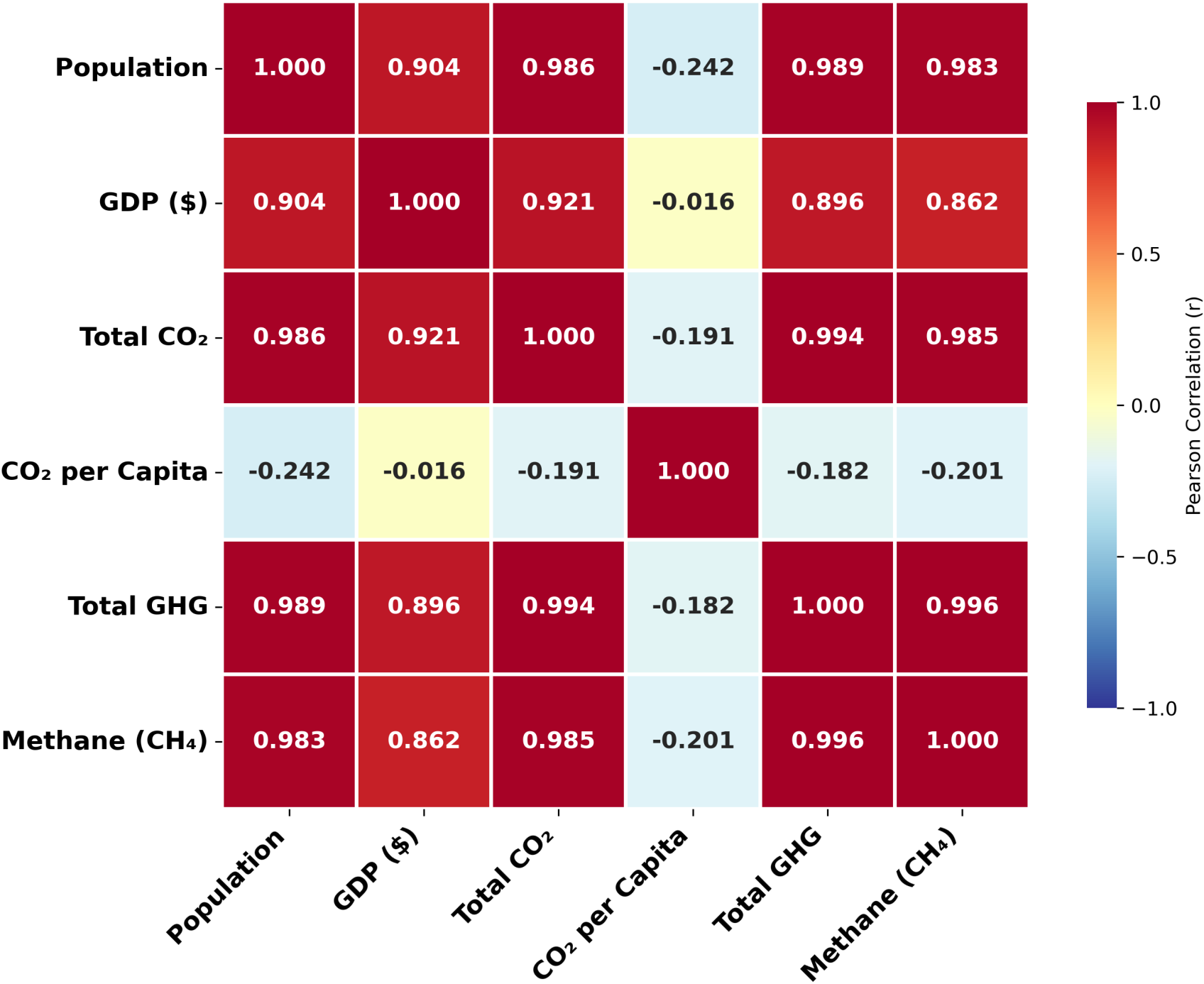
Data

- World-level annual series
- Variables: Population, GDP, total CO₂, CO₂ per capita, GHG, methane

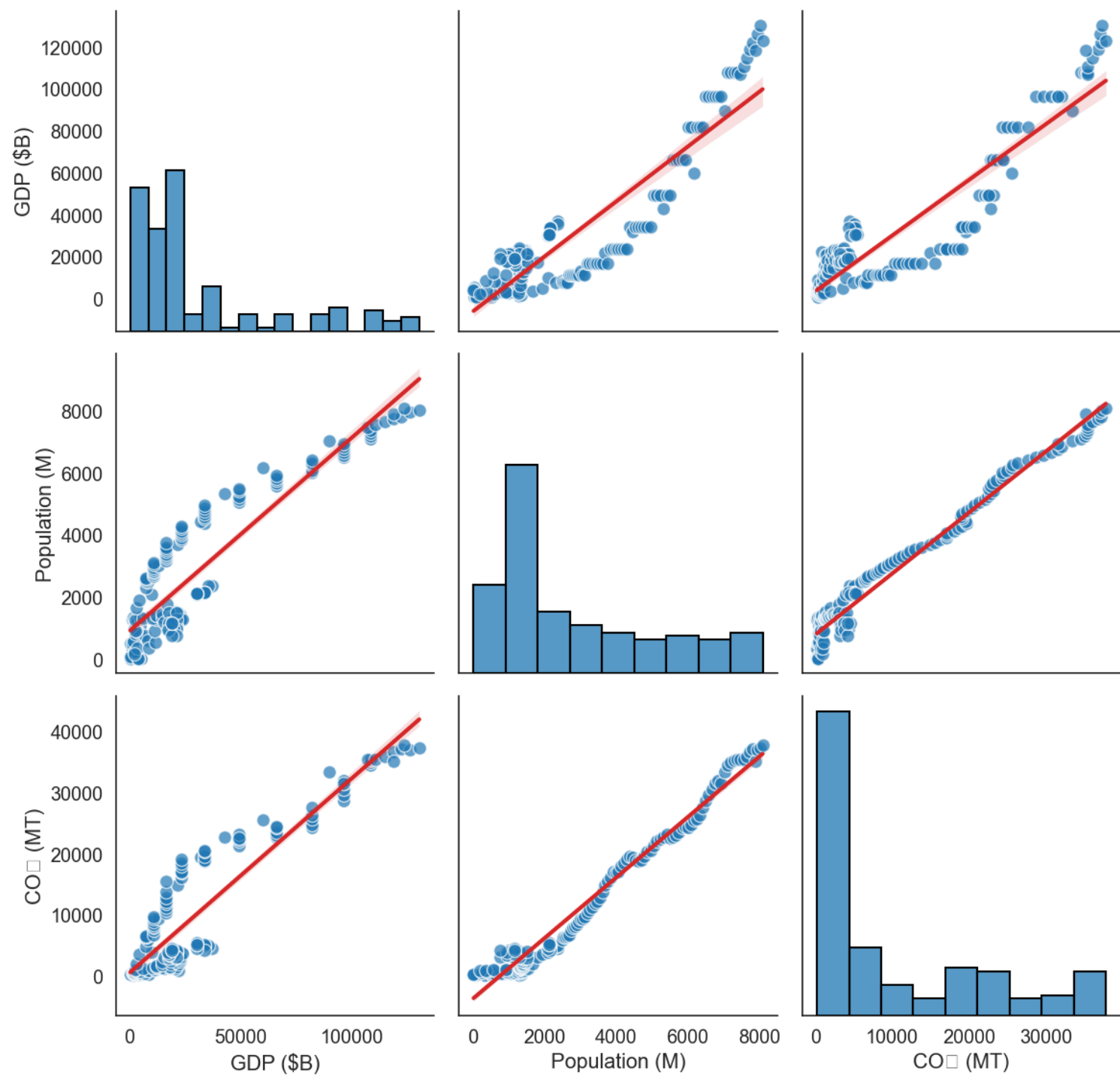
Visuals

- Heatmap: color-coded with r values
- Pairplot: scatter + regression lines

Economic-Environmental Correlation Analysis
Global Emissions & Development Indicators



Economic-Environmental Coupling: GDP, Population & CO₂



Conclusions

Finding	Evidence	Solution
Four Clusters	Small, developed, developing, micro-states	Cluster-specific strategies
Rising Emissions	+5.3% CO ₂ ; stable per capita	Early-warning + efficiency
GDP Coupling	0.97 correlation	3–5% intensity cuts

Thank You

Reference

Dataset by: Shreyansh Dangi, "CO2 Emissions Across Countries, Regions and Sectors," Kaggle, [Online]. Available: <https://www.kaggle.com/datasets/shreyanshdangi/co-emissions-across-countries-regions-and-sectors/data>. [Accessed: 30-Sep-2025]